

Human Brain vs Artificial Neural Network

1. Basic Concept

Aspect	Human Brain	Artificial Neural Network (ANN)
Structure	Made up of biological neurons	Made up of artificial neurons (nodes)
Communication	Signals transmitted via electrical impulses	Signals transmitted via numerical weights
Learning	Learns through experience, repetition	Learns through training using data
Adaptability	Highly flexible and adaptable	Limited to training data and structure

2. Neuron Comparison

Feature	Biological Neuron	Artificial Neuron
Input	Dendrites receive electrical signals	Inputs are numerical data values
Processing Unit	Soma processes signals	Weighted sum + activation function
Output	Axon transmits signal to next neuron	Output is passed to next layer/neuron
Communication	Synapses between neurons	Connections (weights) between nodes

3. Structure & Connectivity

- **Human Brain:**

- Approx. 86 billion neurons
- Complex, dynamic connectivity
- Can rewire (neuroplasticity)

- **Artificial Neural Network:**

- Structured in **layers**:
 - Input layer → Hidden layer(s) → Output layer
 - **Connections have weights**
 - Architecture is **static once defined** (unless retrained)
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4. Learning and Memory

Aspect	Human Brain	Neural Network
Learning Method	Experience, repetition, intuition	Data-driven training (supervised, unsupervised)
Memory	Long-term, short-term, emotional	Stored in weights and biases
Forgetting	Natural forgetting or decay	No forgetting unless weights are reset

5. Energy and Speed

Factor	Human Brain	ANN
Energy Use	Extremely energy-efficient (~20W)	Requires high computation (CPUs/GPUs)
Processing	Parallel, slow in individual neurons	Faster computations using parallel hardware

6. Example Analogy

Task	Human Brain	Neural Network
Image Recognition	Human sees & interprets instantly	Trained CNN identifies objects in images
Language Processing	Natural speaking & understanding	NLP models like ChatGPT understand/generate text